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Large area implementation of bound states in the continuum in a one-dimensional photonic crystal by an optical axis rotating approach

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It is generally accepted that the bound states in the continuum (BICs) are usually realized in a one-dimensional system by the TM-polarized light incident at Brewster's angle. However, the optimal BICs can be achieved only under specific incident light and at a specific polarization incident angle, which poses challenges to its experimental implementation. Serving this purpose, we design a system composed of an anisotropic material (AM) embedded in one-dimensional photonic crystals (1D-PhCs), demonstrating that both TE and TM waves can realize BICs. Moreover, the existence of BICs does not hinge on the specific angle of the incident light. In contrast to the traditional methods of creation system symmetry to achieve symmetry-protected BICs (SP-BICs), we use the 1D-PhC to confine the TE or TM wave in different frequencies first as a form of cavity (localized) mode and rotating the optical axis of AM at a specific angle to realize the field mismatch. Besides, our system can realize Friedrich–Wintgen BICs (FW-BICs) because of the destructive interference of the radiative waves in other specific angles, which can be detected as some Fano resonance collapsing points in reflection spectra. Our results provide more flexible conditions for the realization of BICs under the TE-polarized light, making the implementation of switch Q-factor easier. © 2025 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

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In the continuum, the resonant modes are usually coupled with the radiation channel, resulting in energy leakage [1,2]. However, Neumann and Wigner firstly proved the opposite result of the above wisdom in the quantum field in 1929, called bound states in the continuum (BICs). It is a special state that

contradicts the traditional idea of a lossless resonance mode in an open system [3]. Based on the research of Neumann and Wigner, Friedrich and Wintgen certified BICs can also be generated with the interference between different radiation channels, which was called FW-BIC in 1985 [4]. In 2011, Plotnik *et al.* discovered and analyzed that symmetry-protected BICs (SP-BICs) are caused by the symmetry or the polarization in compatibility of the resonant modes and the propagated modes [5]. BICs as a polarization singularity can realize lossless transmission [6]. Also, when changing the parameters of the system, it can adjust the Q-factor due to the energy decay from the BICs to the radiation mode [7], thus providing a feasible scheme for the future realization of lower threshold lasers [8,9], sensitive biosensors [10], and enhanced quantum optical effects [11,12].

By now, lots of studies realized BICs in nanocavities, some of which can be manipulated by the anisotropic materials (AMs), whose leaky mode can be suppressed by destructive interference in specific optical axis directions [13–18]. Furthermore, multilayer system based on AM is a more controllable and an easily fabricated method for achieving BICs [19–26]. In such a system, one-dimensional photonic crystals are simple and easy to achieve among nanostructures, both in terms of physical mechanism and experimental manufacturing. When consisting of one-dimensional photonic crystal (1D-PhC) and AM, symmetry-protected BICs (SP-BICs) and Friedrich–Wintgen BICs (FW-BICs) can be realized by the TM polarization wave incident at Brewster's angle [19,27,28]. However, BICs are usually limited under the TM polarization of light in the 1D-PhC system, which is rarely supported by the TE-polarized light. In addition, due to limitations in current manufacturing technology, the actual material parameters deviate significantly from simulation results. As a result, Brewster's angle obtained from theoretical calculations cannot be directly applied to experimental conditions, posing numerous challenges for achieving BIC

experimentally. Therefore, the realization of BICs requires not only extending the incident wave to the TE-polarized light in a one-dimensional structure but also relieving Brewster's angle incidence constraint, which is an unsolved problem.

In this Letter, focusing on the band diagram of the one-dimensional photonic crystal (1D-PhC), we demonstrate that the BICs can be achieved not only under the TM-polarized light but also the TE-polarized light. Instead of Brewster's angle incident, which is achieving a full line TM passband also entails achieving a complete TE bandgap, we extend such a line to an area. We design the spatially symmetric system composed of AM embedded into the 1D-PhC, which is designed by two kinds of an artificial material. The PhC constructs an interesting band diagram of different polarizations, i.e., single-polarized light transmission region in different frequency intervals (in this region, only the TE- or only the TM-polarized light can pass through the 1D-PhC), which aim to construct a fairly large range to filter different polarizations first. Then, the SP and FW-BICs as the Fano resonance in the reflection spectra are realized. Compared with the traditional methods of breaking mirror symmetry or spatial inversion symmetry to realize SP-BIC, we propose to use the filtering property of the 1D-PhC instead of the above-mentioned methods and rotating the optical axis of AM to realize the symmetry mismatch between the external excitation modes and eigenmode profiles. The appearance of FW-BICs is due to destructive interference when the optical axis of AM is at a fixed angle. In other words, as long as the single polarization transmission region is within a specific range, BICs can be achieved at any incident angle. In contrast to the previously reported works, our results no longer impose limits on the single-polarized incident light (only the TE or TM wave) and Brewster's angle, providing simpler conditions for the realization of BICs. We construct a symmetric multilayer cavity as shown in Fig. 1. The two sides of the structure are fabricated as 1D-PhCs, which are spatially symmetric, composing of two materials with the thickness d_1 and d_2 and the dielectric

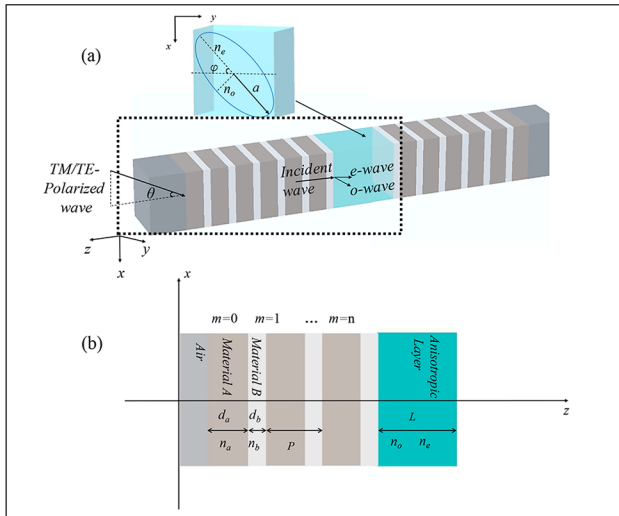


Fig. 1. Multilayer spatially symmetric structure supporting BICs. (a) Infinite 1D PhC with AM defect layer. The refractive index n_a and the thickness d_a of Material A are 1.27 nm and 200 nm, respectively. Material B thickness d_b is 35 nm and $\varepsilon_\infty = 1$; here we assume $\gamma = 0$ and the ω_{pe} set with 3×10^{15} rad/s. Optical properties of the AM are defined by $n_o = 1.52$ and $n_e = 1.72$ and the thickness $L = 9.625$ μm . (b) The side view of (a); P is the period of the 1D-PhC.

constant ε_a and ε_b , respectively. The period of the 1D-PhCs is $P = d_1 + d_2$. Here, ε_b is defined by the Drude model as follows:

$$\varepsilon_b(\omega) = \varepsilon_\infty - \frac{\omega_{pe}^2}{\omega(\omega + i\gamma)}, \quad (1)$$

where ε_∞ is the high-frequency constant, ω is the angular frequency, γ is the collision damping coefficient, and ω_{pe} is the plasma frequency. The AM is placed in the middle of the structure as the defect layer whose thickness is L . Parameters n_o and n_e represent the refractive indices of ordinary and extraordinary light of AM, respectively. The optical axis a of AM is along the y axis on the x - y plane, which can be expressed as follows:

$$a = (\sin \phi, \cos \phi, 0). \quad (2)$$

Here, ϕ is the angle concerning the y axis. The relative permittivity tensor of the AM has the following form:

$$\begin{pmatrix} n_e^2 \sin^2 \phi + n_o^2 \cos^2 \phi & -n_e^2 \sin \phi \cos \phi + n_o^2 \sin \phi \cos \phi & 0 \\ -n_e^2 \sin \phi \cos \phi + n_o^2 \sin \phi \cos \phi & n_e^2 \sin^2 \phi + n_o^2 \cos^2 \phi & 0 \\ 0 & 0 & n_o^2 \end{pmatrix}. \quad (3)$$

When a TE/TM plane wave injects with the angle of θ , the wave vector $\mathbf{k} = (k_x, 0, k_z)$ along the z axis is as follows:

$$k_{iz} = \omega/c\sqrt{\varepsilon_i}\sqrt{\mu_i}\sqrt{1 - \sin^2 \theta/\varepsilon_i\mu_i} \quad (i = a, b), \quad (4)$$

where ε_i and μ_i are the permittivity and the magnetic permeability of Materials A and B, respectively. θ is the angle between the incident wave and the positive direction of the z axis. The Fresnel reflection coefficient of the TE-polarized waves is defined as follows:

$$r_s = \frac{k_{az} - k_{bz}}{k_{az} + k_{bz}}. \quad (5)$$

The TM-wave reflection coefficient can be written as follows:

$$r_p = \frac{\varepsilon_b k_{az} - \varepsilon_a k_{bz}}{\varepsilon_a k_{bz} + \varepsilon_b k_{az}}. \quad (6)$$

The TE dispersion equation in the 1D-PhC satisfies the following:

$$\cos(K\Lambda) = \cos(k_{az}d_a)\cos(k_{bz}d_b) - \frac{1 + r_s^2}{1 - r_s^2} \sin(k_{az}d_a)\sin(k_{bz}d_b). \quad (7)$$

If r_s is replaced by r_p , the TM dispersion equation can be obtained. As shown in Fig. 2(a), the changes of the incident light wavelength and the incident angle of the TE and the TM polarization can be expressed by Eq. (7), which is the band diagram of the 1D-PhC. The red area represents the passing band of the 1D-PhC corresponding to $|\cos(K\Lambda)| < 1$. When $|\cos(K\Lambda)| > 1$, the light cannot propagate in the cyan area. If $|\cos(K\Lambda)| = 1$, the outline depicted by the yellow line is the band boundary. Then, we transform the theoretical band diagram into the momentum space in Fig. 2(b) and plot the light cone line (the red solid line). Above the light cone is the continuum. Here, we observe two interesting regions. Region I is the propagation area of the TM-polarized wave, but the TE-polarized wave is forbidden. Conversely, in Region II, the TE-polarized wave can transmit in the 1D-PhC, while for the TM wave is the bandgap.

Numerical simulation utilizes the finite element method (FEM) [29], where the Bloch boundaries are along the x and y direction. If we change the incident wavelength λ and rotate the optical axis ϕ of AM, BICs exist in the reflection spectra as

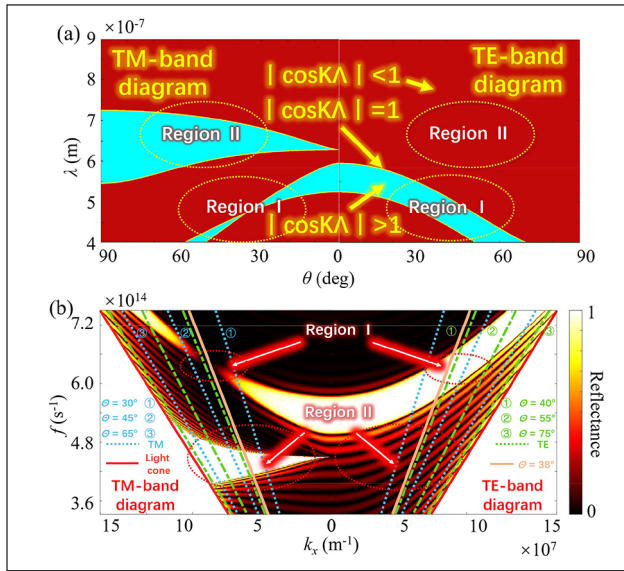


Fig. 2. TM/TE-polarized wave band distribution (left/right) of one-dimensional photonic crystals. (a) Red region indicates the transmission region of the 1D-PhC corresponding to $|\cos(K\lambda)| < 1$; the cyan area is the bandgap for the light corresponding to $|\cos(K\lambda)| > 1$, and the yellow line is the boundary of the band corresponding to $|\cos(K\lambda)| = 1$. (b) Band structure in momentum space is calculated by the transfer matrix method; the 1D-PhCs are assumed to be 15 pairs.

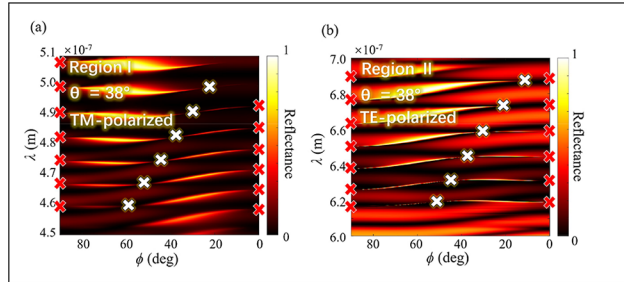


Fig. 3. Illustration of BICs injected by different polarized light in the reflectance spectra. The incident angle here set by 38° (the solid orange line in Fig. 2(b)). (a) BICs occur in Region I illuminated by the TM-polarized wave. (b) BICs occur in Region II illuminated by the TE-polarized wave. The white crosses correspond to the FW-BICs, and the red crosses correspond to the SP-BICs.

shown in Fig. 3 with different colored crosses, which are Fano resonance collapses. In the symmetrical system, the 1D-PhC acts as a Fabry–Perot cavity to confine the TE or the TM wave. In other words, for the first frequency domain, the 1D-PhC can be regarded as a perfect mirror for the TE wave, preventing the propagation of the TM wave. Thus, the TE-localized model can be decoupled from the TM continuum.

As shown in Fig. 3(a), the AM can be considered as isotropic at $\phi = 0^\circ$ when the TM wave incidents into the structure where the TE wave couples only with the e wave, but the o wave propagating in the AM is excited by the TM wave. Thus, the TE mode cannot contribute to the TM continuum and the SP-BICs occur, which we have plotted as the red crosses. The process of decoupling the localized state of a TM from the TE-polarized continuum at Region II is the same as above. As shown in

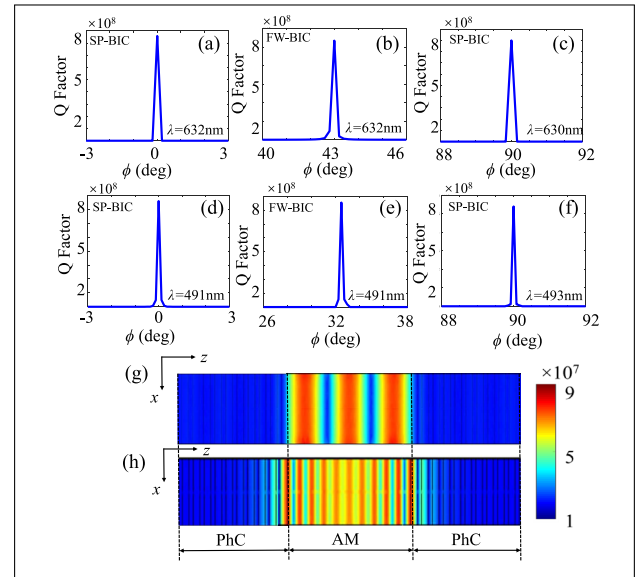


Fig. 4. Illustration of SP and FW-BICs for some of the Q-factor in the reflectance spectra. (a), (c), (d), and (f) are the Q-factor of the SP-BICs when $\lambda = 632$ nm, 630 nm for the TE polarization and 491 nm and 493 nm for the TM polarization, respectively. (b) and (e) are the Q-factor of the FW-BICs when $\lambda = 632$ nm for the TE polarization and 491 nm for the TM polarization, respectively. (g) and (h) are the electric field distribution $|E|$ of the eigenmodes of SP-BICs and the FW-BICs, respectively.

Fig. 3(b), when TE-polarized wave is incident into the structure, the frequency of Region II of the 1D-PhC is in the TM bandgap rather than in the TE passband region, leading to the appearance of SP-BICs due to the decoupling of the TM wave from the TE continuum.

Furthermore, FW-BICs are observed in our system when the incident light is at a specific angle ϕ from 0° to 90° in Regions I and II. This kind of BICs appears due to the full vector interference of o and e wave with the help of the radiation channel. At this time, the TM wave inside the 1D-PhC cannot couple with the localized mode in AM. Therefore, we use the band characteristics of the 1D-PhC to achieve BICs by different polarized light at different frequencies.

Then, we calculate the quality factor and the electric field distribution of the system eigenmodes to prove that these Fano collapses are BICs in Fig. 4. Figures 4(a)–4(c) are the case of the TE wave incident in Fig. 3(b), and Fig. 4(d)–4(f) are the case of the TM wave incident in Fig. 3(c). The Q-factor is much higher than other modes, both SP-BICs and FW-BICs occur, corresponding to the BICs in Fig. 3. When SP- and FW-BICs occur, the energy is confined in the AM in Figs. 4(g) and 4(h), respectively.

Finally, we aim to prove Brewster's angle is not the necessary condition for the BICs in Fig. 5. In Region I of Fig. 2(b), we choose different incident angles at 30° , 45° , and 65° with the TM polarization, and in Region II, we choose different incident angles at 40° , 55° , and 75° with the TE polarization. The white crosses are the FW-BICs and the red crosses are the SP-BICs in the reflectance spectra, showing us that as long as it is in the single-polarized light band passing region, the incident angle range in such polarized light passband region to achieve BICs can be very large. In other words, the implementation

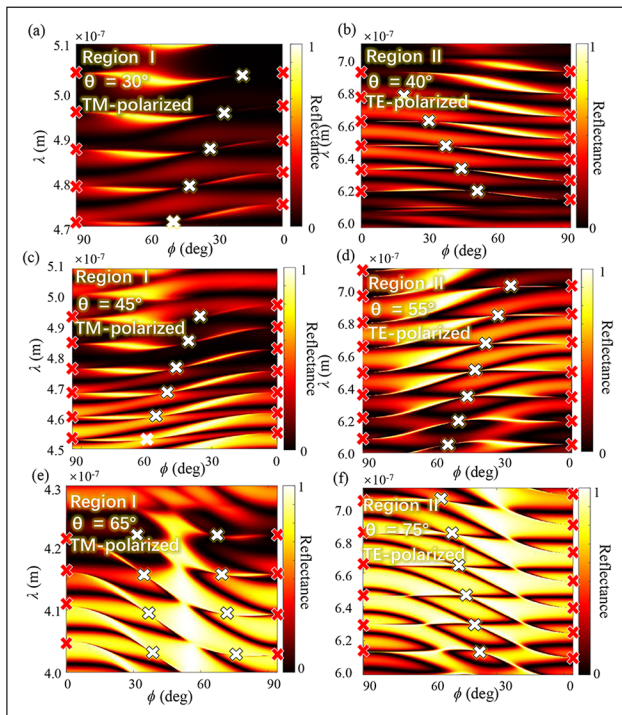


Fig. 5. Illustration of BICs injected by different polarized light in the reflectance spectra. (a), (c), and (e) BICs occur in Region I illuminated by the TM-polarized wave. The incident angle θ is set at 30° , 45° , and 65° , respectively. (b), (d), and (f) BICs occur in Region II illuminated by the TE-polarized wave. The incident angle θ are set at 40° , 55° , and 75° , respectively. The white crosses correspond to the FW-BICs and the red crosses correspond to the SP-BICs.

of BIC is not limited to a fixed incidence angle, and both TE and TM waves can realize two kinds of BICs in a wide range angle.

In conclusion, compared with breaking symmetry to achieve SP-BICs in the past, we propose to use the band structure of 1D PhCs to filter a TE or TM wave firstly, which aim to achieve mode mismatch between the internal mode and external mode. Then, by rotating the optical axis of the anisotropic material, we demonstrate the spatially symmetric system can support both SP-BICs and FW-BICs. These BICs can be observed as the Fano resonance in the reflectance spectra, which is the incident wave frequency in the range of single polarization transmitted area. In addition, our study also shows that as long as there is frequency of the incident wave in such areas, Brewster's angle is not required anymore, and instead, these two kinds of BICs can be achieved by incident at an arbitrary angle. We provide a more flexible method for the experimental implementation of BICs with lower error, fewer restrictions,

and easier implementation, which facilitates the application of BICs.

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Data availability. Data underlying the results presented in this Letter are not publicly available at this time but may be obtained from the authors upon reasonable request.

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